

Positioning Coherence Metrics for Low-Rank Models in 2025

Theoretical Development Team

2025

Introduction

The Challenge

- ▶ Evaluation of unsupervised low-rank models remains difficult
- ▶ Practitioners rely on reconstruction error and explained variance
- ▶ These metrics fail to capture semantic quality
- ▶ Joshua Cook's coherence project addresses this critical gap

Overview

- ▶ The evaluation crisis in unsupervised learning
- ▶ Evolution of coherence metrics beyond topic modeling
- ▶ Computational breakthroughs in mutual information
- ▶ Industry adoption and research gaps
- ▶ Strategic positioning of the coherence project

The Evaluation Crisis

Fundamental Barriers

- ▶ **Kleinberg's impossibility theorem:** Clustering limitations
- ▶ **Locatello's work:** Disentangled representations
- ▶ Hard limits on perfect unsupervised evaluation
- ▶ Need for pragmatic multi-criteria approaches

Current Evaluation Methods

Categories

- ▶ **Cross-validation:** Alex Williams' "speckled holdout"
- ▶ **Stability metrics:** Reproducibility under perturbations
- ▶ **Information-theoretic:** Quantify preserved information

Limitations

- ▶ Reconstruction error ignores semantics
- ▶ Stability metrics computationally expensive
- ▶ Information measures struggle with high dimensions

Evolution of Coherence Metrics

Recent Advances (2020+)

- ▶ **Contextualized Topic Coherence (CTC)**
 - ▶ Leverages large language models
 - ▶ Evaluates semantic coherence
 - ▶ Beyond statistical co-occurrence

Theoretical Connections

- ▶ VAEs pursue PCA-like directions
- ▶ Linear autoencoders span same subspace as PCA
- ▶ Mathematical connections suggest generalizability
- ▶ Yet explicit frameworks remain absent

Research Gap

- ▶ No published work by Joshua Cook found
- ▶ Novel research addressing identified gap
- ▶ Natural extension of topic modeling principles
- ▶ Mutual information between loading matrix features

Mutual Information Breakthroughs

Computational Advances

- ▶ **MINE Framework:** Neural MI estimation
 - ▶ Linear scaling with dimensionality
 - ▶ Handles 1000+ dimensions
- ▶ **Variational bounds:** Bias-variance trade-offs
- ▶ **Latent MI approximation:** Exploits low-dimensional structure

Advantages for Evaluation

- ▶ Captures linear and nonlinear dependencies
- ▶ Invariant to monotonic transformations
- ▶ Principled information-theoretic interpretation
- ▶ Better correlation with downstream performance

Theoretical Justification

- ▶ Measures statistical dependence without functional forms
- ▶ Suitable for diverse low-rank models
- ▶ Captures complex relationships
- ▶ Challenge: Accurate estimation for high dimensions

Alternative Evaluation Approaches

Methodological Diversity

Stability-Based

- ▶ Consistency under perturbations
- ▶ Robustness guarantees
- ▶ High computational cost

Geometric/Topological

- ▶ Persistent homology
- ▶ Chamfer distance
- ▶ Structural properties

Downstream Tasks

- ▶ Gold standard evaluation
- ▶ Requires labeled data
- ▶ May miss representation qualities

Key Finding

- ▶ **No universal best metric**
- ▶ Performance depends on:
 - ▶ Data characteristics
 - ▶ Evaluation goals
- ▶ Ensemble approaches recommended

Validation Methodologies

- ▶ Human judgment correlation studies
- ▶ Synthetic data experiments
- ▶ Meta-evaluation frameworks
- ▶ Comprehensive evaluation needed

Industry Adoption

Current State

Dominant Tools

- ▶ **Scikit-learn:** Basic metrics only
 - ▶ Reconstruction error
 - ▶ Explained variance
- ▶ Limited specialized libraries:
 - ▶ scikit-mdr
 - ▶ DR quality

Industry Challenges

- ▶ Billion-scale dataset scalability
- ▶ Real-time evaluation constraints
- ▶ Limited knowledge transfer
- ▶ Minimal public documentation

Academic vs Industry

Academic Usage

- ▶ More diverse metrics
- ▶ Local/global structure preservation
- ▶ Parameter sensitivity
- ▶ Domain-specific advances

Persistent Problems

- ▶ Overreliance on reconstruction error
- ▶ Insufficient robustness testing
- ▶ Research-practice divide

Open Research Questions

Fundamental Questions

Semantic Preservation

- ▶ How to evaluate meaningful relationships?
- ▶ Current approaches miss human-recognizable patterns

Multi-Scale Structure

- ▶ Hierarchical data representations
- ▶ Scale-dependent evaluation

Causal Relationships

- ▶ Dimensionality reduction may destroy causality
- ▶ Preserving correlations insufficient

Computational Challenges

- ▶ Quadratic scaling with dataset size
- ▶ Streaming evaluation unexplored
- ▶ Need approximate methods with guarantees

Domain-Specific Needs

- ▶ **Time series:** Temporal-aware metrics
- ▶ **Multi-modal:** Cross-source evaluation
- ▶ **Healthcare:** Clinical relevance
- ▶ Each domain has unique requirements

Positioning the Coherence Project

Addressing Multiple Gaps

- ▶ Extends semantic coherence to general low-rank models
- ▶ Alternative to reconstruction-focused evaluation
- ▶ Leverages ML computational advances
- ▶ Direct interpretability through loading matrices

Project Strengths

Theoretical

- ▶ Principled foundation
- ▶ Information-theoretic rigor
- ▶ General applicability

Practical

- ▶ Computationally feasible
- ▶ Industry-relevant
- ▶ Bridges research-practice gap

Potential Challenges

- ▶ Scalability for very high dimensions
- ▶ Validation against existing metrics
- ▶ Establishing advantages
- ▶ Domain-specific adaptations

Strategic Recommendations

Development Priorities

1. **Efficient Implementation**

- ▶ Neural estimators for high dimensions
- ▶ k-NN methods for moderate scales
- ▶ Leverage modern ML techniques

2. **Empirical Validation**

- ▶ Compare against established metrics
- ▶ Diverse datasets and tasks
- ▶ Human judgment correlation

3. **Domain Adaptations**

- ▶ Time series specialization
- ▶ Multi-modal extensions
- ▶ Streaming data support

Collaboration Opportunities

Industry Partners

- ▶ Large-scale validation
- ▶ Practical feedback
- ▶ Real-world deployment

Academic Partners

- ▶ Theoretical foundations
- ▶ Related work connections
- ▶ Rigorous evaluation

Open Source

- ▶ Community contributions
- ▶ Accelerated adoption
- ▶ Transparency

Future Impact

Potential Contributions

▶ **For Practitioners**

- ▶ Interpretable evaluation beyond reconstruction
- ▶ Practical tools for model assessment
- ▶ Clear implementation guidelines

▶ **For Research**

- ▶ New evaluation standards
- ▶ Bridge theory-practice gap
- ▶ Enable principled development

Broader Significance

- ▶ Understanding what models learn
- ▶ Moving beyond statistical measures
- ▶ Toward semantic understanding
- ▶ Critical for complex models/datasets

Conclusion

Key Messages

- ▶ Evaluation crisis requires new approaches
- ▶ Coherence metrics offer promising direction
- ▶ Computational advances enable implementation
- ▶ Significant gaps remain to be addressed

The Opportunity

- ▶ Joshua Cook's project positioned to:
 - ▶ Provide crucial evaluation framework
 - ▶ Bridge academic-industry divide
 - ▶ Enable better model development
 - ▶ Advance the field significantly

Call to Action

- ▶ Develop efficient implementations
- ▶ Validate comprehensively
- ▶ Collaborate broadly
- ▶ Create lasting impact